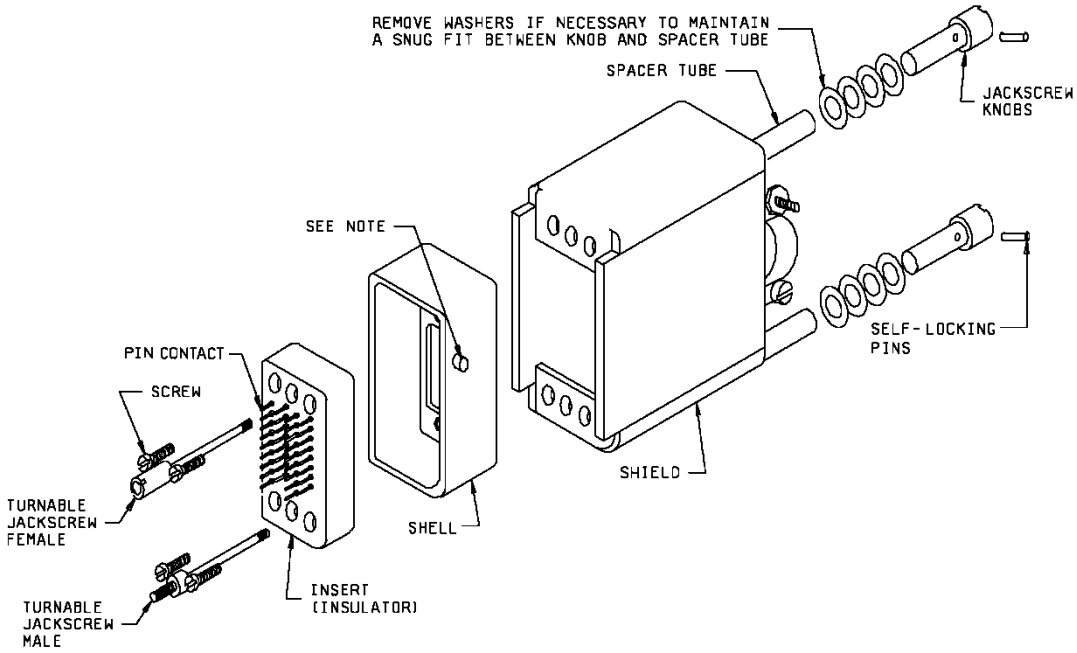
INCH-POUND
MIL-DTL-28748/3F
w/AMENDMENT 1
8 August 2014
SUPERSEDING
MIL-DTL-28748/3F
9 September 2004

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, RECTANGULAR, RACK AND PANEL, CRIMP TYPE REMOVABLE PIN CONTACTS, SIZES 16 AND 20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.




Connector

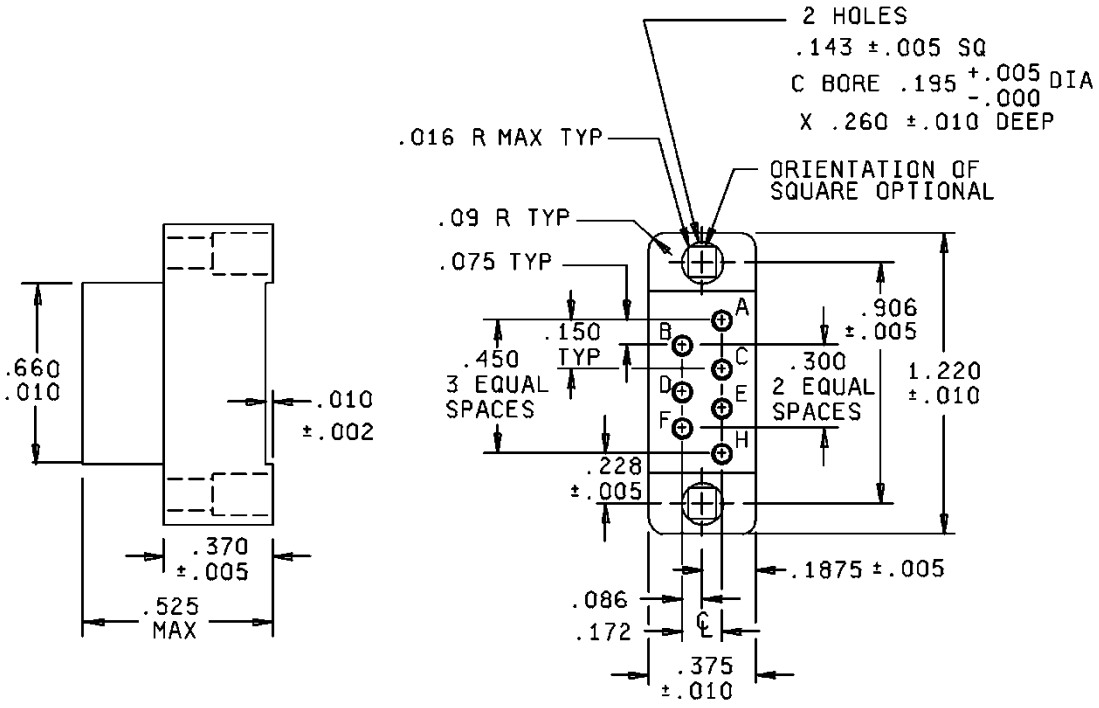
NOTE: The polarization feature (pin on the plug shell, groove on the receptacle shell) shall be adjacent to the side closest to the number 1 or A contact position.

Configuration A

FIGURE 1. Plug connector with shield, plug shell, and turnable jackscrews.

AMSC N/A FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .002 | 0.05 | .143 | 3.63 | .300 | 7.62 |
| .005 | 0.13 | .150 | 3.81 | .370 | 9.40 |
| .010 | 0.25 | .172 | 4.37 | .375 | 9.53 |
| .016 | 0.41 | .1875 | 4.765 | .450 | 11.43 |
| .075 | 1.91 | .195 | 4.95 | .525 | 13.34 |
| .086 | 2.18 | .228 | 5.79 | .660 | 16.76 |
| .09 | 2.3 | .260 | 6.60 | .906 | 23.01 |
| | | | | 1.220 | 30.99 |

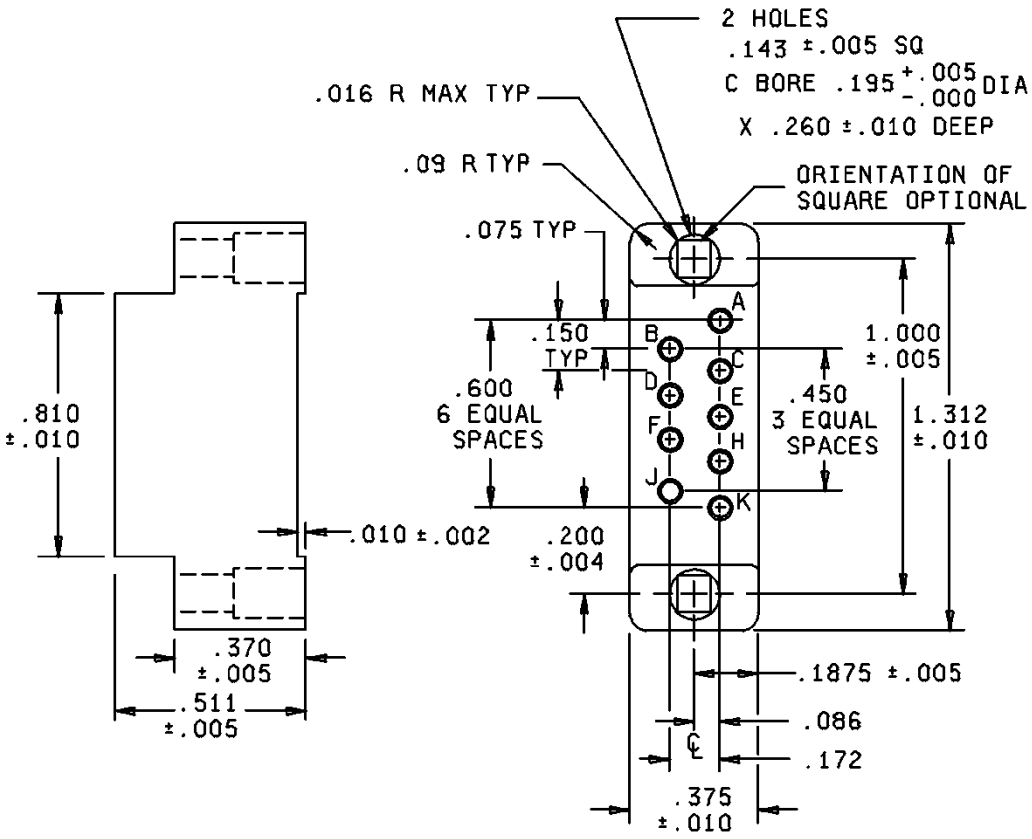
Insert with 7 contacts

Insert designator A

FIGURE 2. Insert arrangements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .002 | 0.05 | .143 | 3.63 | .375 | 9.53 |
| .004 | 0.10 | .150 | 3.81 | .450 | 11.43 |
| .005 | 0.13 | .172 | 4.37 | .511 | 12.98 |
| .010 | 0.25 | .1875 | 4.76 | .600 | 15.24 |
| .016 | 0.41 | .195 | 4.95 | .810 | 20.57 |
| .075 | 1.91 | .200 | 5.08 | 1.000 | 25.40 |
| .086 | 2.18 | .260 | 6.60 | 1.312 | 33.32 |
| .09 | 2.3 | .370 | 9.40 | | |

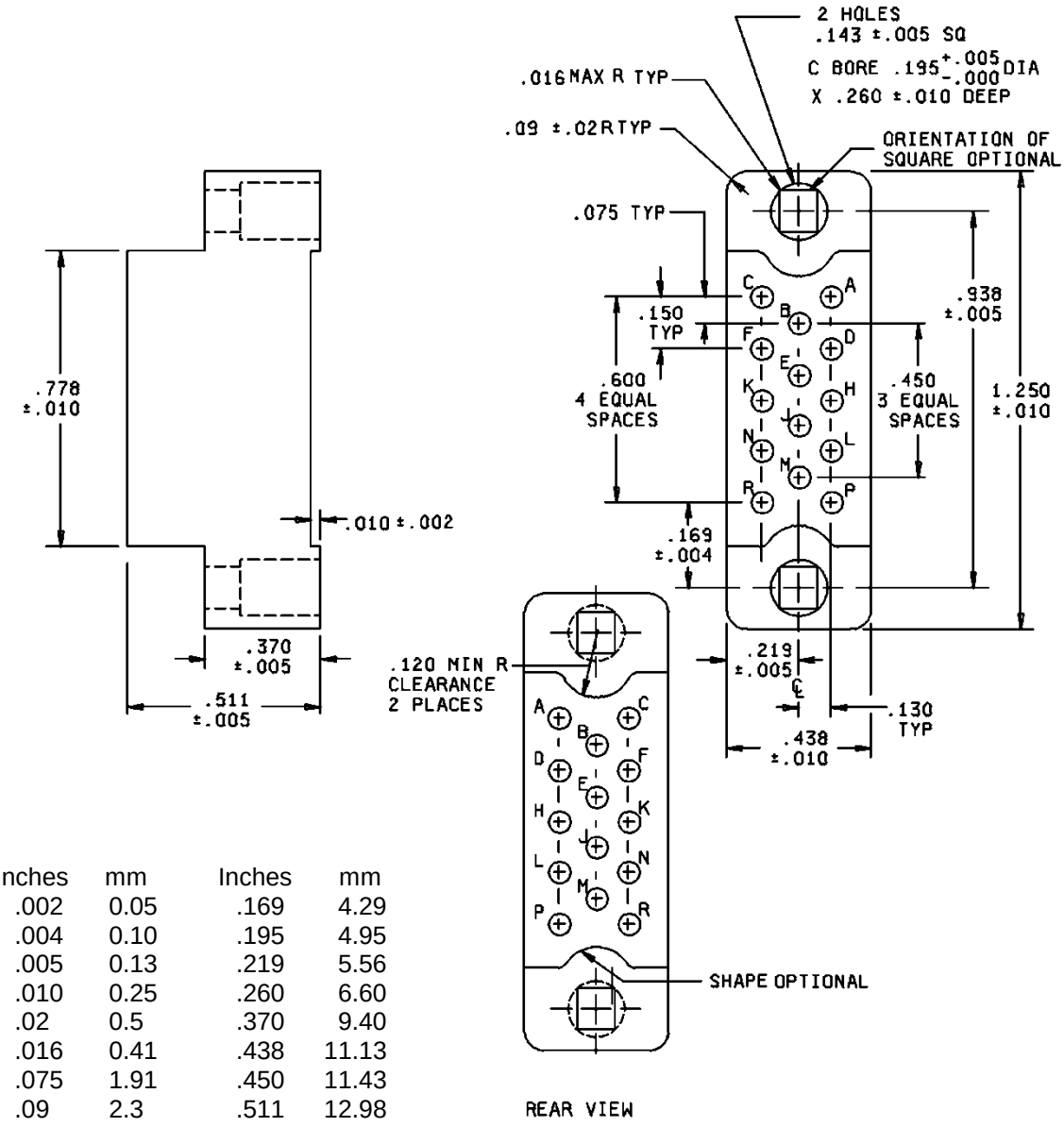
Insert with 9 contacts

Insert designator B

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .002 | 0.05 | .169 | 4.29 |
| .004 | 0.10 | .195 | 4.95 |
| .005 | 0.13 | .219 | 5.56 |
| .010 | 0.25 | .260 | 6.60 |
| .02 | 0.5 | .370 | 9.40 |
| .016 | 0.41 | .438 | 11.13 |
| .075 | 1.91 | .450 | 11.43 |
| .09 | 2.3 | .511 | 12.98 |
| .120 | 3.06 | .600 | 15.24 |
| .130 | 3.30 | .778 | 19.76 |
| .143 | 3.63 | .938 | 23.83 |
| .150 | 3.81 | 1.250 | 31.75 |

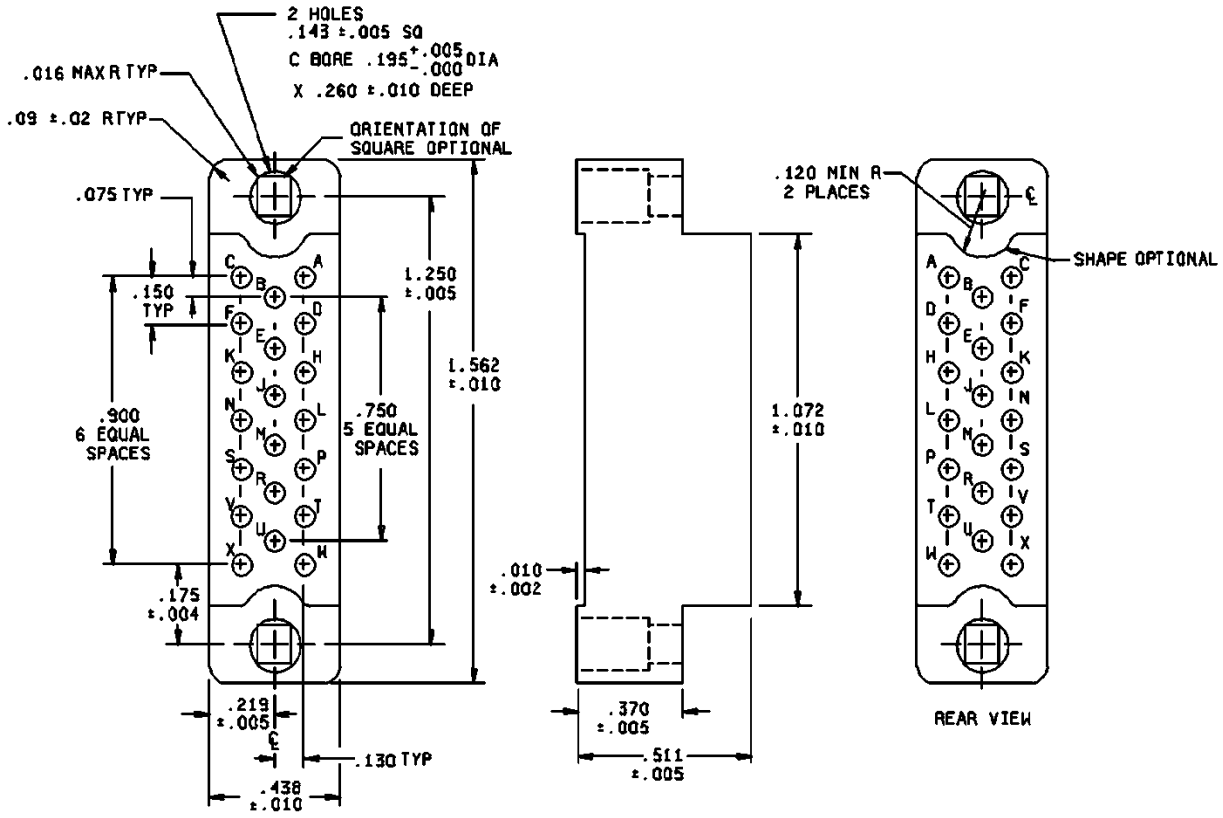
Insert with 14 contacts

Insert designator C

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .002 | 0.05 | .120 | 3.05 | .370 | 9.40 |
| .004 | 0.10 | .130 | 3.30 | .438 | 11.13 |
| .005 | 0.13 | .143 | 3.63 | .511 | 12.98 |
| .010 | 0.25 | .150 | 3.81 | .750 | 19.05 |
| .016 | 0.41 | .175 | 4.45 | .900 | 22.86 |
| .02 | 0.5 | .195 | 4.95 | 1.072 | 27.23 |
| .075 | 1.91 | .219 | 5.56 | 1.250 | 31.75 |
| .09 | 2.3 | .260 | 6.60 | 1.562 | 39.67 |
| | | | | 2.290 | 58.17 |

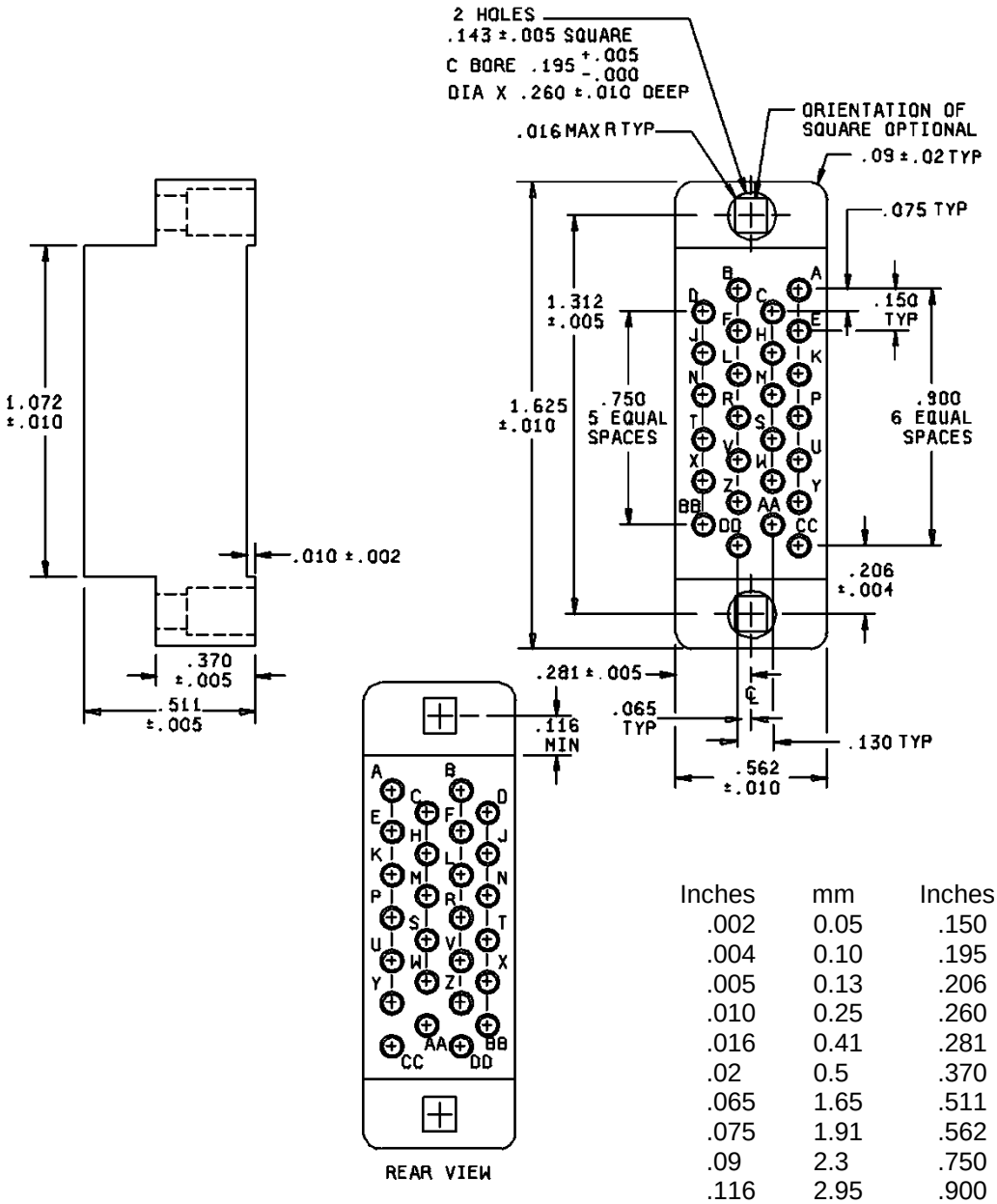
Insert with 20 contacts

Insert designator D

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .002 | 0.05 | .150 | 3.81 |
| .004 | 0.10 | .195 | 4.95 |
| .005 | 0.13 | .206 | 5.23 |
| .010 | 0.25 | .260 | 6.60 |
| .016 | 0.41 | .281 | 7.14 |
| .02 | 0.5 | .370 | 9.40 |
| .065 | 1.65 | .511 | 12.98 |
| .075 | 1.91 | .562 | 14.28 |
| .09 | 2.3 | .750 | 19.05 |
| .116 | 2.95 | .900 | 22.86 |
| .143 | 3.63 | 1.072 | 27.23 |
| .130 | 3.30 | 1.312 | 33.32 |
| .143 | 3.63 | 1.625 | 41.28 |

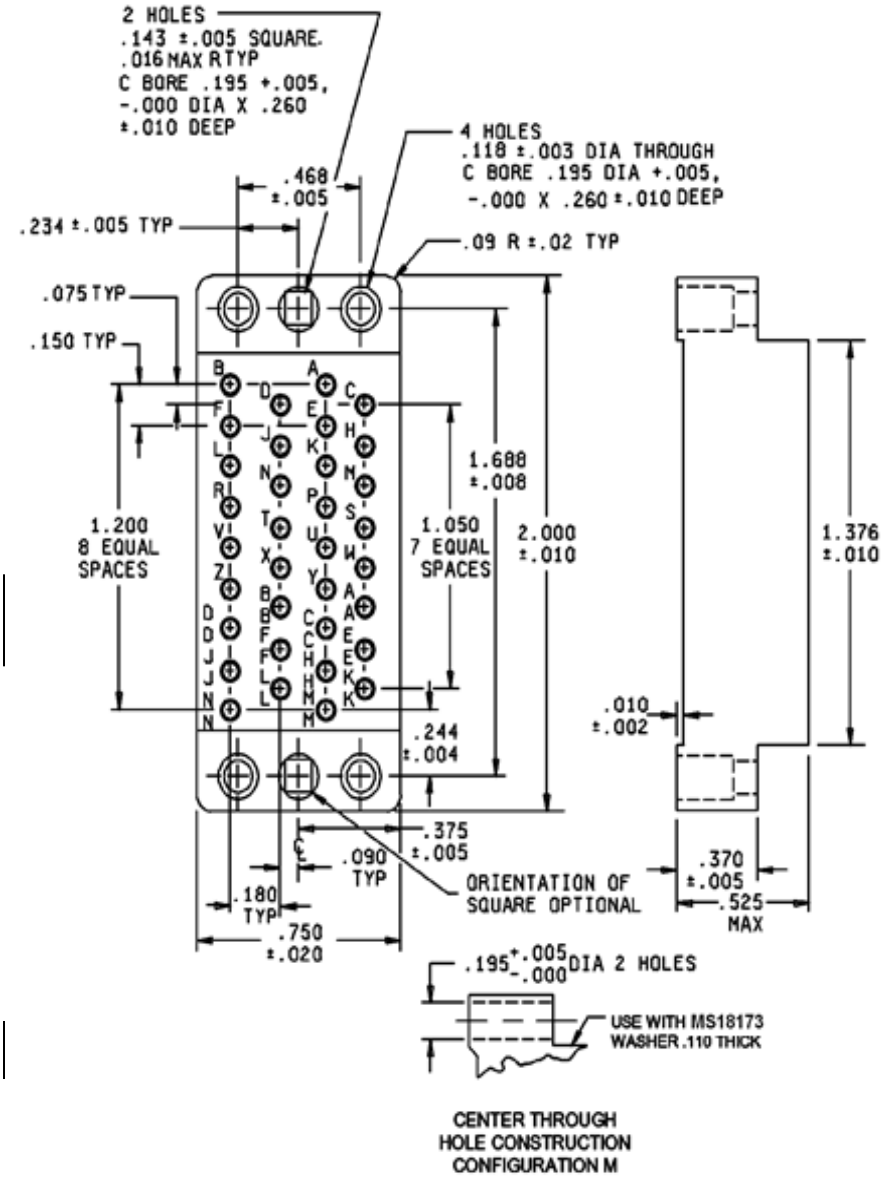
Insert with 26 contacts

Insert designator E

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .004 | 0.10 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .02 | 0.5 |
| .020 | .051 |
| .075 | 1.91 |
| .09 | 2.3 |
| .090 | 2.29 |
| .110 | 2.79 |
| .118 | 3.00 |
| .143 | 3.63 |
| .150 | 3.81 |
| .180 | 4.57 |
| .195 | 4.95 |
| .234 | 5.94 |
| .244 | 6.20 |
| .260 | 6.60 |
| .370 | 9.40 |
| .375 | 9.53 |
| .468 | 11.88 |
| .525 | 13.34 |
| .750 | 19.05 |
| 1.050 | 26.67 |
| 1.200 | 30.48 |
| 1.376 | 34.95 |
| 1.688 | 42.88 |
| 2.000 | 50.80 |

Configuration F - Suitable for all applications, except for one-piece jackscrews.
Configuration M - For long jackscrews and shield only (through hole alternate construction).

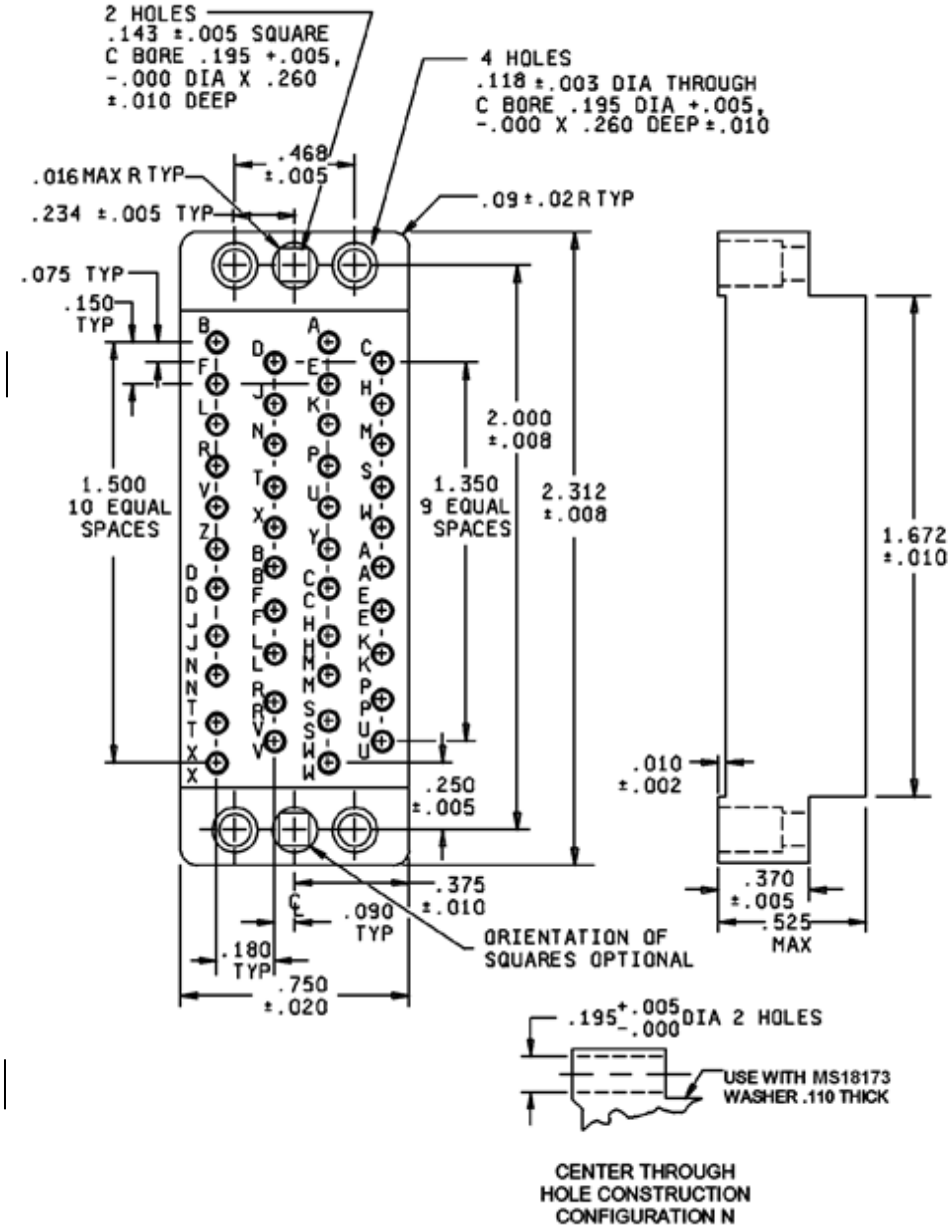
Insert with 34 contacts

Insert designators F and M

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .02 | 0.5 |
| .020 | 0.51 |
| .075 | 1.91 |
| .09 | 2.3 |
| .090 | 2.29 |
| .110 | 2.79 |
| .118 | 3.00 |
| .143 | 3.63 |
| .150 | 3.81 |
| .180 | 4.57 |
| .195 | 4.95 |
| .234 | 5.94 |
| .250 | 6.35 |
| .260 | 6.60 |
| .370 | 9.40 |
| .375 | 9.53 |
| .468 | 11.89 |
| .525 | 13.34 |
| .750 | 19.05 |
| 1.350 | 34.29 |
| 1.500 | 38.10 |
| 1.672 | 42.47 |
| 2.000 | 50.80 |
| 2.312 | 58.72 |

Configuration G - Suitable for all applications, except one-piece jackscrews.
Configuration N - For long jackscrews and shield only (through hole alternate construction).

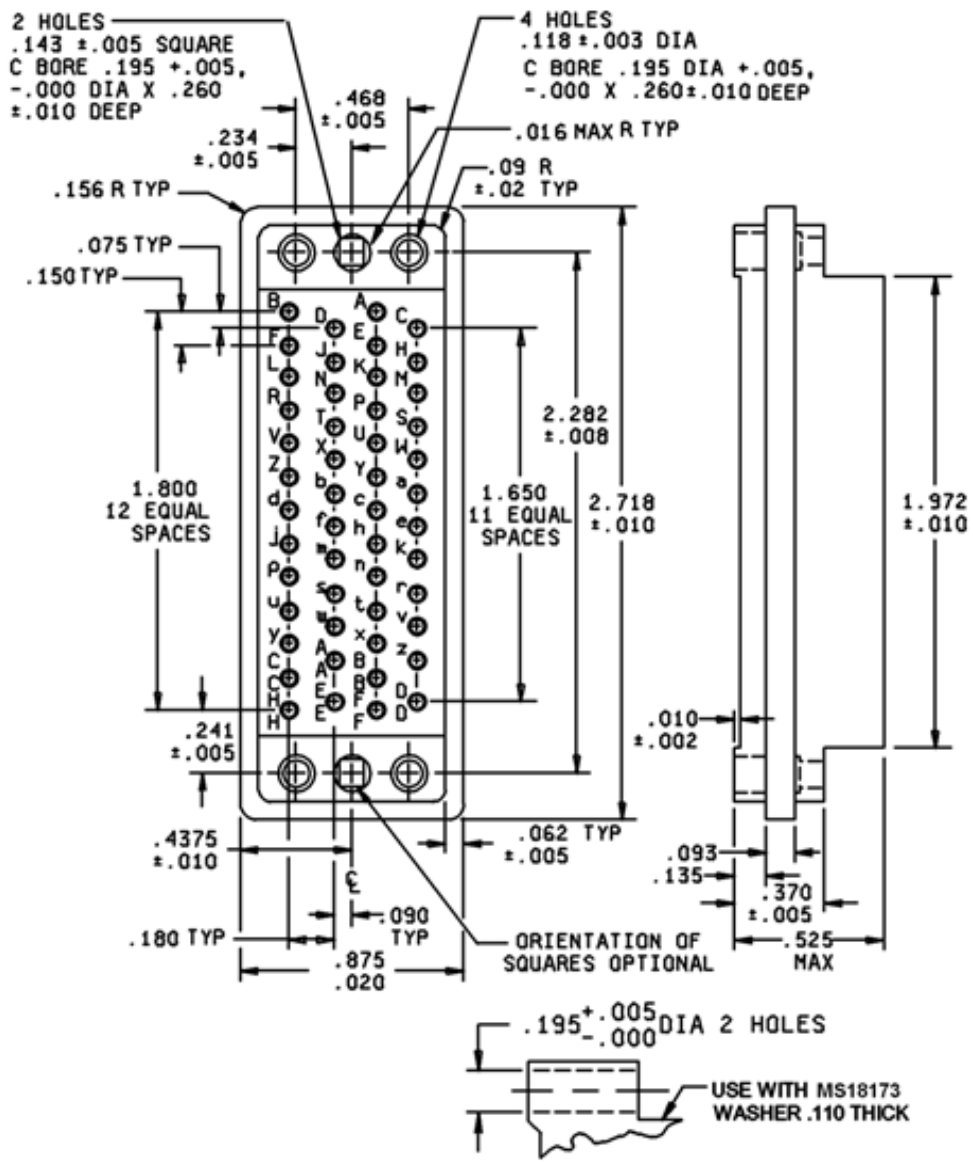
Insert with 42 contacts

Insert designators G and N

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




CENTER THROUGH
HOLE CONSTRUCTION
CONFIGURATION P

| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .02 | 0.5 |
| .020 | 0.51 |
| .062 | 1.57 |
| .075 | 1.91 |
| .09 | 2.3 |
| .090 | 2.29 |
| .093 | 2.36 |
| .110 | 2.79 |
| .118 | 3.00 |
| .135 | 3.43 |
| .143 | 3.63 |
| .150 | 3.81 |
| .156 | 3.96 |
| .180 | 4.57 |
| .195 | 4.95 |
| .234 | 5.94 |
| .241 | 6.12 |
| .260 | 6.60 |
| .370 | 9.40 |
| .4375 | 11.11 |
| .468 | 11.89 |
| .525 | 12.98 |
| .875 | 22.23 |
| 1.650 | 41.91 |
| 1.800 | 45.72 |
| 1.972 | 50.09 |
| 2.282 | 57.96 |
| 2.718 | 69.04 |

Configuration H - Suitable for all applications, except one-piece jackscrews.
Configuration P - For long jackscrews and shield only (through hole alternate construction).

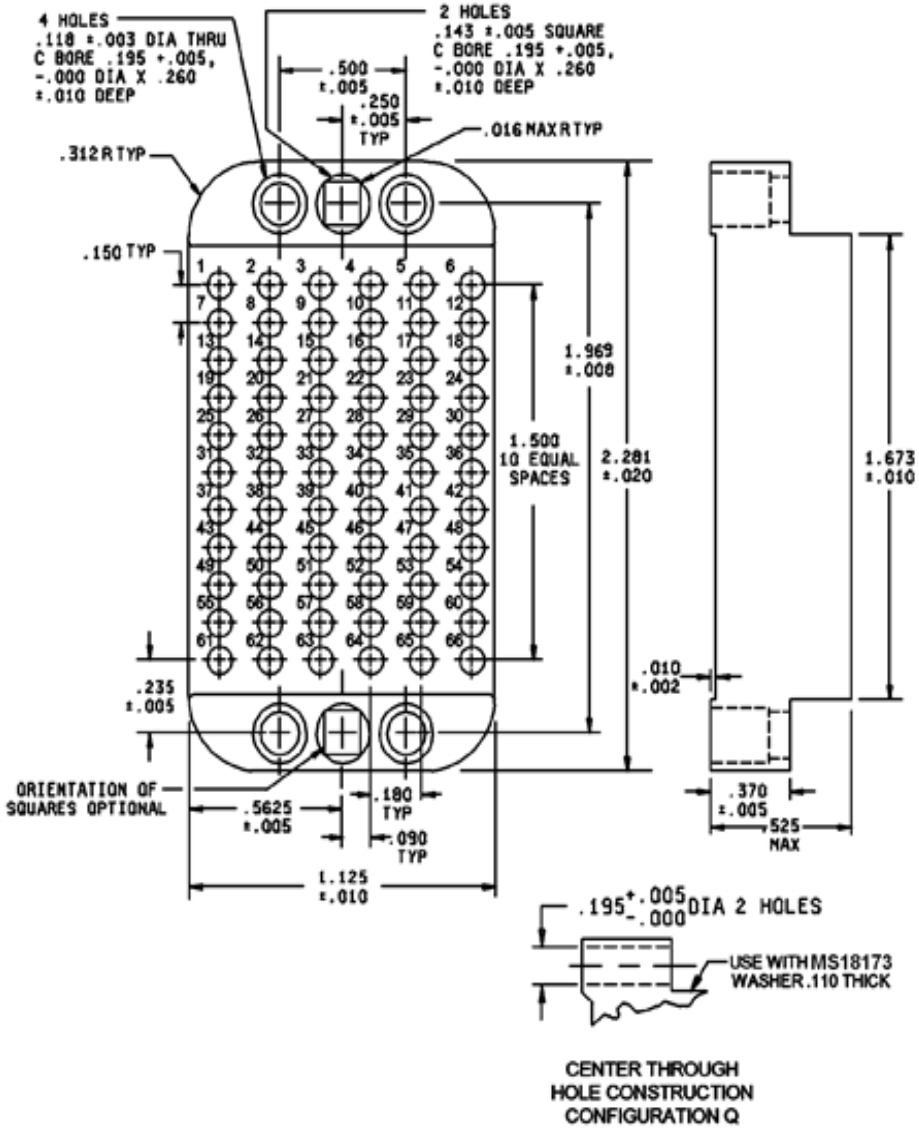
Insert with 50 contacts

Insert designators H and P

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .020 | 0.51 |
| .090 | 2.29 |
| .110 | 2.79 |
| .118 | 3.00 |
| .143 | 3.63 |
| .150 | 3.81 |
| .180 | 4.57 |
| .195 | 4.95 |
| .235 | 5.97 |
| .250 | 6.35 |
| .260 | 6.60 |
| .312 | 7.92 |
| .370 | 9.40 |
| .500 | 12.70 |
| .525 | 13.34 |
| .5625 | 14.288 |
| 1.125 | 28.58 |
| 1.500 | 38.10 |
| 1.673 | 42.49 |
| 1.969 | 50.01 |
| 2.281 | 57.94 |

Configuration J - Suitable for all applications, except for one-piece jackscrews.
Configuration Q - For long jackscrews and shield only (through hole alternate construction).

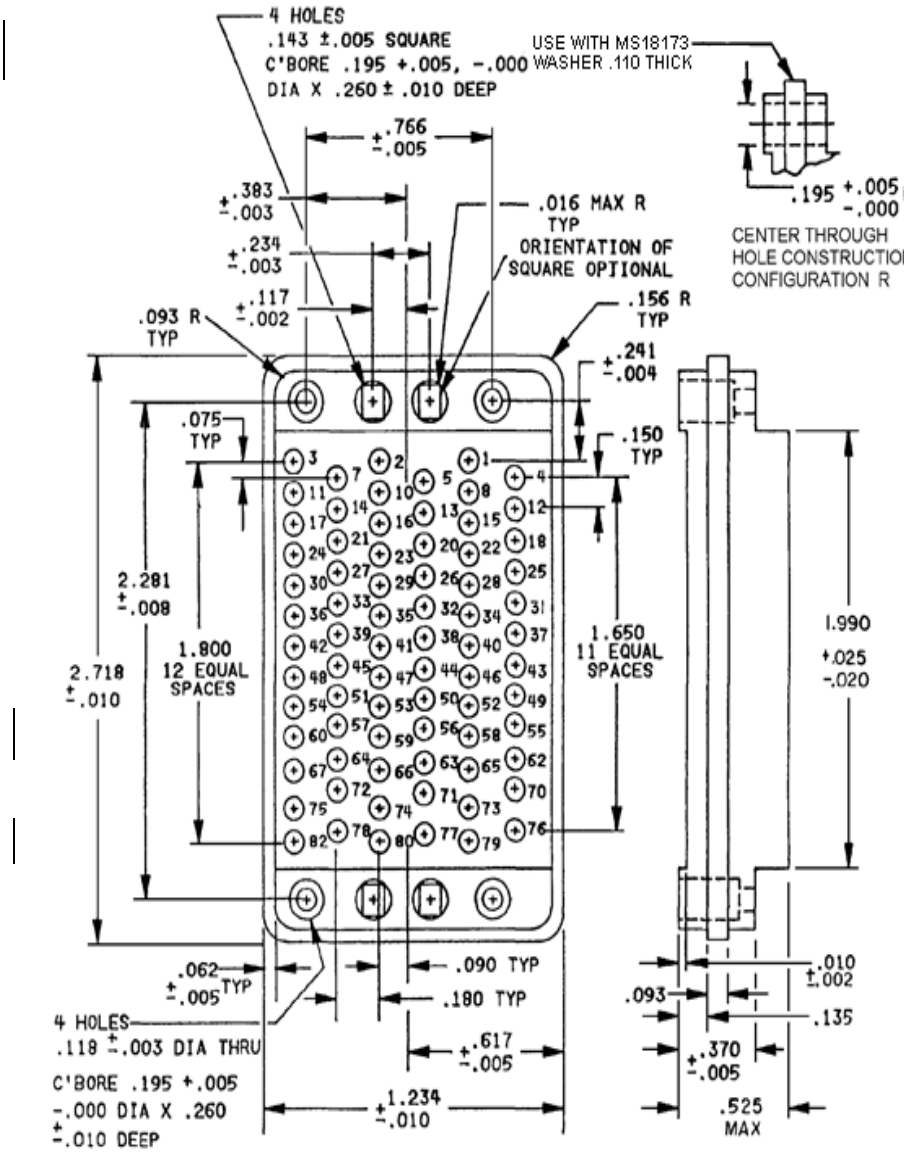
Insert with 66 contacts

Insert designators J and Q

FIGURE 2. Insert arrangements - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:11Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .004 | 0.10 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .020 | 0.51 |
| .025 | 0.64 |
| .062 | 1.57 |
| .075 | 1.91 |
| .090 | 2.29 |
| .093 | 2.36 |
| .110 | 2.79 |
| .117 | 2.97 |
| .118 | 3.00 |
| .135 | 3.43 |
| .143 | 3.63 |
| .150 | 3.81 |
| .156 | 3.96 |
| .180 | 4.57 |
| .195 | 4.95 |
| .234 | 5.94 |
| .241 | 6.12 |
| .260 | 6.60 |
| .370 | 9.40 |
| .383 | 9.73 |
| .525 | 13.34 |
| .617 | 15.67 |
| .766 | 19.46 |
| 1.234 | 31.34 |
| 1.650 | 41.91 |
| 1.800 | 45.72 |
| 1.990 | 50.55 |
| 2.281 | 57.94 |
| 2.718 | 69.04 |

Configuration K - Suitable for all applications, except one-piece jackscrews.
Configuration R - For long jackscrews and shield only (through hole alternate construction).

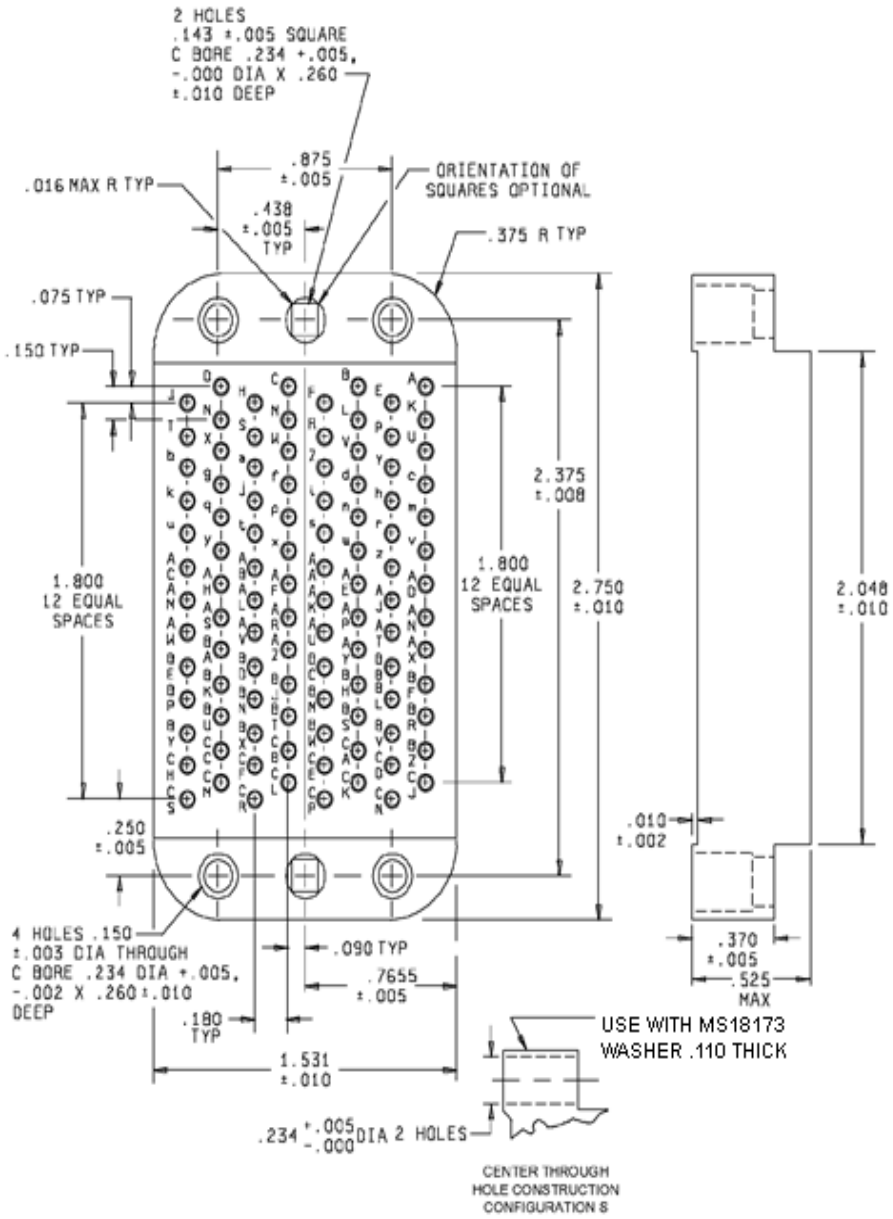
Insert with 75 contacts

Insert designators K and R

FIGURE 2. Insert arrangements - Continued.



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Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .016 | 0.41 |
| .075 | 1.91 |
| .090 | 2.29 |
| .110 | 2.79 |
| .143 | 3.63 |
| .150 | 3.81 |
| .180 | 4.57 |
| .234 | 5.94 |
| .250 | 6.35 |
| .260 | 6.60 |
| .370 | 9.40 |
| .375 | 9.53 |
| .438 | 11.13 |
| .525 | 13.34 |
| .7655 | 19.44 |
| .875 | 22.23 |
| 1.531 | 38.89 |
| 1.800 | 45.72 |
| 2.048 | 52.02 |
| 2.375 | 60.33 |
| 2.750 | 69.85 |

Configurations L - Suitable for all applications, except one-piece jackscrews.
Configurations S - For long jackscrews and shield only (through hole alternate construction).

Insert with 104 contacts

Insert designators L and S

FIGURE 2. Insert arrangements - Continued.



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REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Insert material: See MIL-DTL-28748.

Mating connector: See MIL-DTL-28748/2 and MIL-DTL-28748/4.

Part or Identifying Number (PIN): As shown in the following example and table I: 1/

See table I for permissible PINs:

(See PIN codes)

| M28748/3 – | A | 1 | H | L | 1 A |
|---|---|---|---|---|---|
| Basic number of specification sheet | Insert designator | Shield and shield clamp location or retaining plate | Shell and polarization position | Jackscrews or guidepins | Contacts (see table II) |

PIN codes:

| Insert designator (See table III) | Number of contact positions |
|---|---|
| A | 7 |
| B | 9 |
| C | 14 |
| D | 20 |
| E | 26 |
| F | 34 |
| G | 42 |
| H | 50 |
| J | 66 |
| K | 75 |
| L | 104 |
| Insert designator through hole 5/ | |
| M | 34 |
| N | 42 |
| P | 50 |
| Q | 66 |
| R | 75 |
| S | 104 |

Shield and shield clamp location or retaining plate 2/

0 – No shield or retaining plate

Shield and shield clamp location

1 - top MS18192
2 -side MS18192
3 - top MS18193
4 -side MS18193

Retaining plate

5 – MS18198
6 – MS18199
7 – MS18200

Shell and polarization designators 3/

0 - No shell included

Plug shell

A – A of MS18203
B – B of MS18203
C – C of MS18203
D – D of MS18203
E – E of MS18203
F – F of MS18203
G – G of MS18203
H - Unpolarized MS18203

Receptacle shell

J – A of MS18204
K– B of MS18204
L– C of MS18204
M– D of MS18204
N– E of MS18204
P– F of MS18204
Q– G of MS18204
R– Unpolarized MS18204

Jackscrews or guidepins 4/

0 – No Jackscrews or guidepins.
F - Fixed jackscrews, MS18196 (not used with shield).
G - Guidepins, MS18197 (not used with shield).
L - Long jackscrews MS18194
S - Short jackscrews MS18195 (not used with shield).

Jack and guide location: The jackscrew female (turning or fixed) or guide socket shall be located by the A or 1 pin contact.

See footnotes on next page.



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1/ The number zero (0) is used to indicate which parts are not included.
2/ Retaining plate and shells are not available for insert size A.
3/ Polarization positions are for location of plug and receptacle shell pins and slots.
4/ Jackscrew designator L: Multi-piece jackscrews will be supplied with insert designators A through L. Long one-piece jackscrews will be supplied for insert designators M through S.
5/ Inserts M through S have through holes for use with long one-piece jackscrew designs only.
6/ Mold bodies and hardware from different manufacturers are not necessarily interchangeable.

TABLE I. Permissible PIN's (connector assembly).

<table>
<tr><td>Configuration</td><td colspan="5">Accessories</td></tr>
<tr><td>Receptacle</td><td>Insert</td><td>Shield and clamp position/ or retaining plate</td><td>Shell and polarization position</td><td>Jackscrews</td><td>Contacts</td></tr>
<tr><td rowspan="4">M28748/3-</td><td>A</td><td>0</td><td>0</td><td rowspan="4">0, F, G</td><td rowspan="11">See table II</td></tr>
<tr><td>B<br>C<br>D<br>E</td><td>0<br>Retaining Plate 5</td><td rowspan="3">0,<br>J thru R</td></tr>
<tr><td>F<br>G<br>H<br>K</td><td>0<br>Retaining Plate 6</td></tr>
<tr><td>J<br>L</td><td>0<br>Retaining Plate 7</td></tr>
<tr><td>Plug</td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="7">M28748/3-</td><td>A</td><td>0</td><td rowspan="7">0,<br>A thru H</td><td rowspan="7">0, L, G, S</td></tr>
<tr><td>B<br>C<br>D<br>E</td><td rowspan="3">0<br>Shield 1, 2</td></tr>
<tr><td>F<br>G<br>H<br>K</td><td>0</td></tr>
<tr><td>J<br>L</td><td>0<br>Shield 3, 4</td></tr>
<tr><td>M<br>N<br>P<br>R</td><td>0<br>Shield 1, 2</td></tr>
<tr><td>Q<br>S</td><td>0<br>Shield 3, 4</td></tr>
</table>



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TABLE II. Contact size percent 1/

| Contact | Sizes 16-16 M39029/34-273 (percent) | Sizes 16-20 M39029/34-272 (percent) | Sizes 20-20 M39029/34-271 (percent) |
|---|---|---|---|
| 1A | 100 | 0 | 0 |
| 1B | 90 | 0 | 0 |
| 1C | 80 | 0 | 0 |
| 1D | 70 | 0 | 0 |
| 1E | 60 | 0 | 0 |
| 1F | 50 | 0 | 0 |
| 1G | 40 | 0 | 0 |
| 1H | 30 | 0 | 0 |
| 1J | 20 | 0 | 0 |
| 1K | 10 | 0 | 0 |
| 1L | 0 | 0 | 0 |
| 2A | 0 | 100 | 0 |
| 2B | 0 | 90 | 0 |
| 2C | 0 | 80 | 0 |
| 2D | 0 | 70 | 0 |
| 2E | 0 | 60 | 0 |
| 2F | 0 | 50 | 0 |
| 2G | 0 | 40 | 0 |
| 2H | 0 | 30 | 0 |
| 2J | 0 | 20 | 0 |
| 2K | 0 | 10 | 0 |
| 3A | 0 | 0 | 100 |
| 3B | 0 | 0 | 90 |
| 3C | 0 | 0 | 80 |
| 3D | 0 | 0 | 70 |
| 3E | 0 | 0 | 60 |
| 3F | 0 | 0 | 50 |
| 3G | 0 | 0 | 40 |
| 3H | 0 | 0 | 30 |
| 3J | 0 | 0 | 20 |
| 3K | 0 | 0 | 10 |

1/ Contact percent size = number of contact positions times the percent in table II. Example: 34 X .80 = 27.2 or 27 contacts supplied. Values of .5 and above are rounded to the next contact.

Crimp tool M22520/1-01.
Positioner M22520/1-03 (positioner A color blue/red).
Installation tool M81969/18-01.
Removal tool M81969/20-01.



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For superseded insert Military Standard sheet PIN's to M28748 PIN's see table III.

TABLE III. Supersession data (inserts).

| MS PIN | M28748/3 PIN | Number of contact positions |
|---|---|---|
| MS14010-1 | M28748/3-A0001L | 7 |
| MS14007-1 | M28748/3-B0001L | 9 |
| MS18174-1 | M28748/3-C0001L | 14 |
| MS18176-1 | M28748/3-D0001L | 20 |
| MS14008-1 | M28748/3-E0001L | 26 |
| MS18178-1 | M28748/3-F0001L | 34 |
| MS18180-1 | M28748/3-G0001L | 42 |
| MS18182-1 | M28748/3-H0001L | 50 |
| MS18184-1 | M28748/3-J0001L | 66 |
| MS18187-1 | M28748/3-K0001L | 75 |
| MS18189-1 | M28748/3-L0001L | 104 |
| MS18178-2 | M28748/3-M0001L | 34 1/ |
| MS18180-2 | M28748/3-N0001L | 42 1/ |
| MS18182-2 | M28748/3-P0001L | 50 1/ |
| MS18184-2 | M28748/3-Q0001L | 66 1/ |
| MS18187-2 | M28748/3-R0001L | 75 1/ |
| MS18189-2 | M28748/3-S0001L | 104 1/ |

1/ Insert through hole designator. Inserts –2 (M through S) have through holes for use with long one-piece jackscrew designs only.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference documents. In addition to MIL-DTL-28748, this document references the following:

MIL-DTL-28748/2
MIL-DTL-28748/4
MS18173
MS18192
MS18193
MS18194
MS18195
MS18196
MS18197
MS18198
MS18199
MS18200
MS18203
MS18204



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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2014-035)

Review activities:
Army - AT, MI
Navy - AS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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